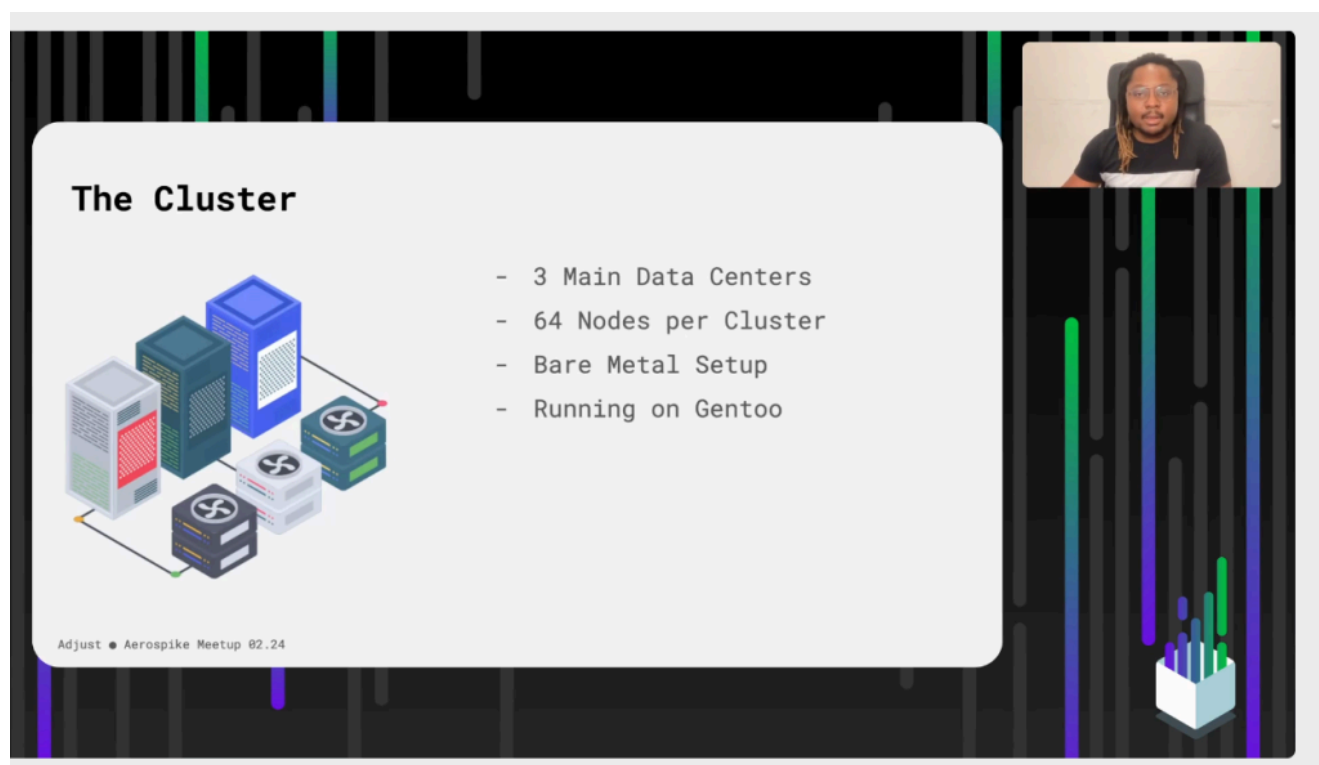


ДАННЫЕ / ПОТОКОВАЯ ПЕРЕДАЧА ДАННЫХ / БАЗЫ ДАННЫХ

Как управлять 45 миллиардами записей клиентов с помощью Aerospike

На прошлой неделе на саммите Aerospike по работе с данными в режиме реального времени Бубуньо Ньяво из Adjust рассказал, как компания использовала Aerospike, чтобы помочь клиентам отслеживать окупаемость инвестиций в маркетинговые каналы.

1 июля 2024 года, 6:51 [Джоаб Джексон](#)



Изображения с конференции Aerospike Real-Time Data Summit.

Что делать, если ваши операции выходят за рамки возможностей одной базы данных?

Для берлинского поставщика услуг мобильных измерений **Adjust** решением стал **Aerospike** — высокопроизводительное хранилище данных NoSQL в режиме реального времени, которое можно использовать в нескольких центрах обработки данных.

как компания использовала Aerospike, чтобы помочь клиентам отслеживать окупаемость маркетинговых каналов.

Сервис Adjust может генерировать в среднем 52 миллиона запросов в минуту. Эти запросы могут привести к необходимости выполнения какой-либо операции, например запроса, и, конечно же, к необходимости согласования состояния. Клиент может опубликовать материал в Meta, LinkedIn или другой социальной сети, а Adjust подсчитает количество людей, просмотревших контент, и количество переходов по ссылкам.

«В зависимости от того, какая это операция, мы извлекаем одни данные, записываем другие. Иногда мы записываем данные пакетами, иногда удаляем их, а затем возвращаем ответ на эти запросы», — сказал Ньявор.



Aerospike — это база данных реального времени, созданная для неограниченного масштабирования, высокой скорости и экономии. Наши клиенты готовы к новым вызовам благодаря платформе данных с самой низкой задержкой и самой высокой пропускной способностью. Мы предоставляем облачные технологии и искусственный интеллект таким ведущим организациям, как Adobe, Airtel, Criteo, Experian и PayPal.

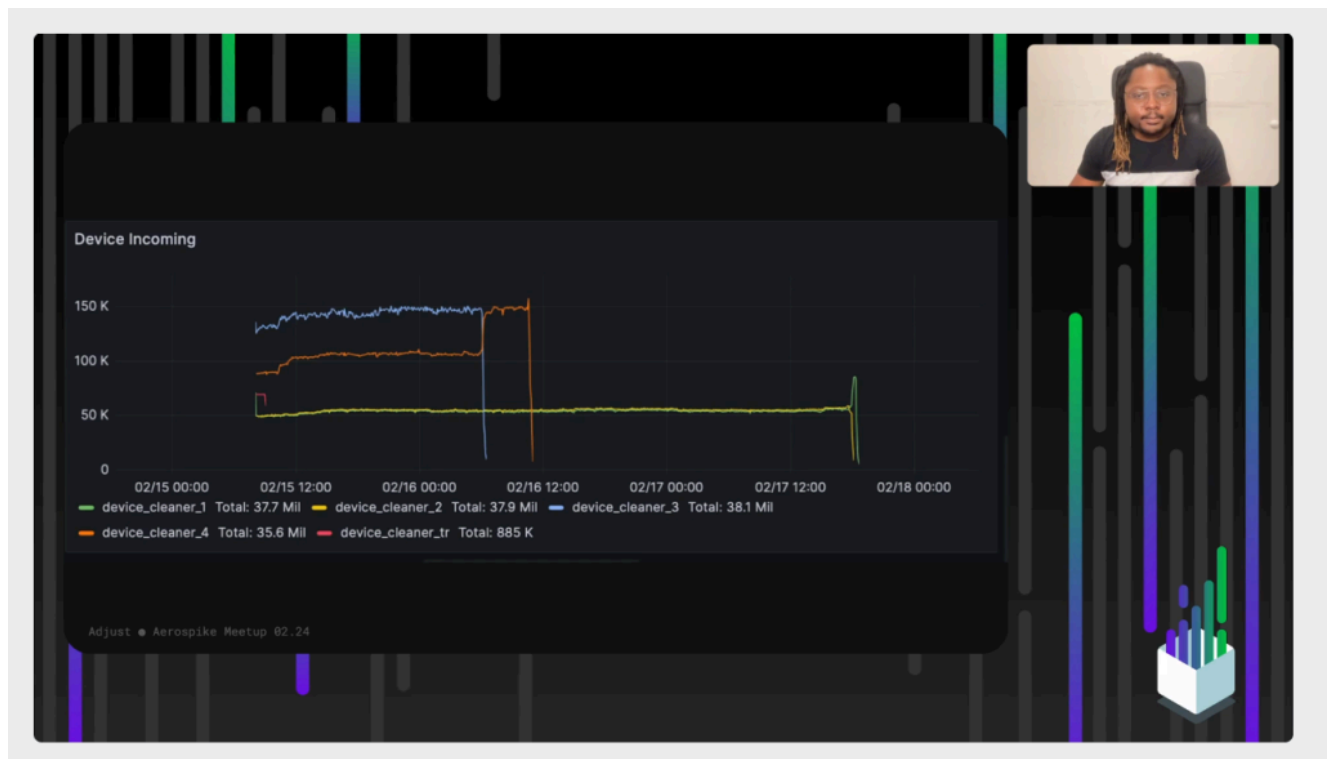
[Узнать больше →](#)

В целом компания хранит около 45 миллиардов записей в **Aerospike**, и это только данные о состоянии устройств. При среднем размере записи в 512 байт объем данных составляет 351 ТБ.

The data is stored in three separate three separate clusters, located in geographically-dispersed data centers. Each cluster has 64 nodes and runs on bare metal, with **Gentoo Linux** serving as the operating system. Each server has about 400GB of RAM and 16TB of solid-state of **NVMe disk space**, and a 10 Gigabit network card. Either two or three copies of the data are kept as backup.

1. Timescale Hyperfunctions Ease PostgreSQL Query Complexities
2. Maria DB Gets Reactive with a Non-Blocking Connector for Java
3. 6 Tips for Better Data Science in the Cloud
4. Distributed SQL Takes Databases to the Next Level
5. Alation Delivers a New Data Governance App

“So that if single rack goes offline, it doesn’t send us into a tailspin,” Nyavor said.



A chart showing the average number of devices connecting to Aerospike.

Zoom

Beyond Key-Store Values

The Aerospike key-value store **was launched** in 2009 (originally as CitrusLeaf) and quickly found an audience in the online advertising industry for storing and subsequently analyzing customer **cookies at rapid speed**.

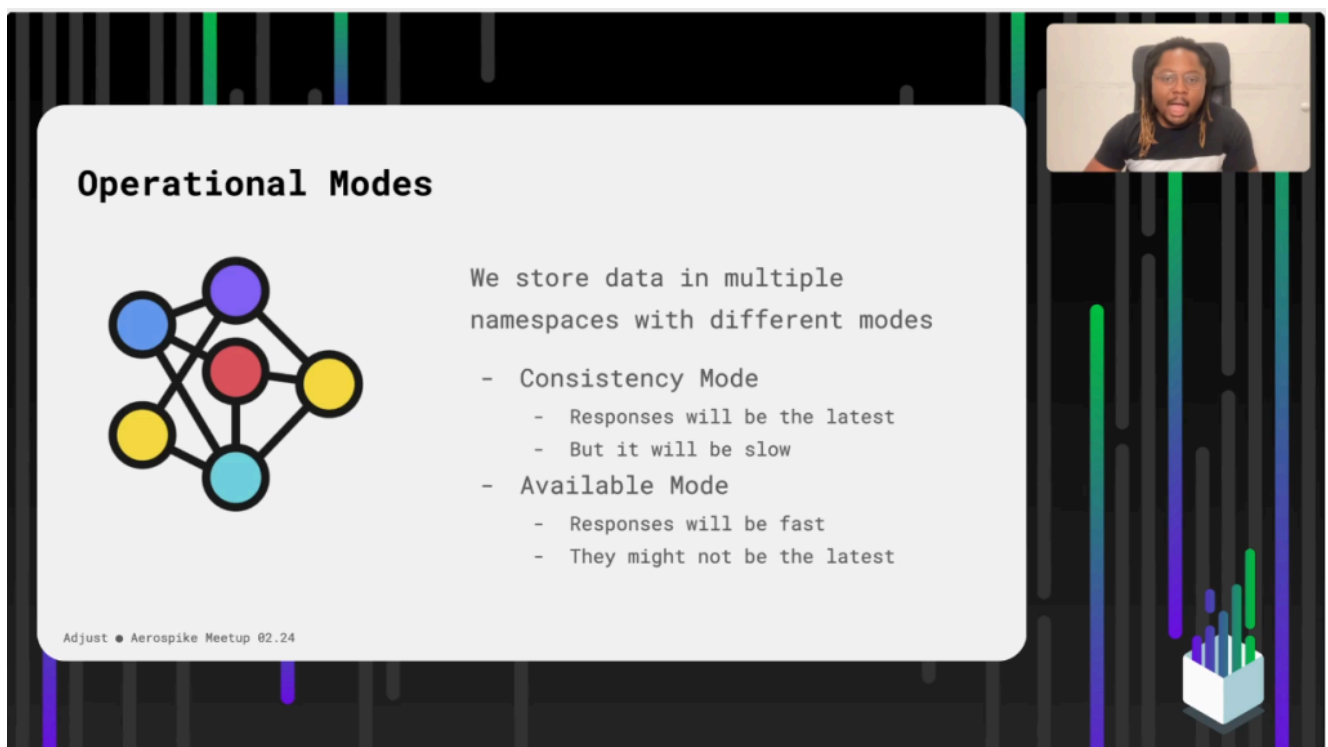
Subsequent releases **expanded the analytics, incorporated batch processing**, and introduced **secondary indexes and cross-data center replication**.

which can help online shops easily build out recommendation systems.

For Adjust, low latency was critical. Customers wanted data updated as close to real-time as possible. This is a challenge given the cross-cluster communications.

As with any distributed system with duplicate data, Adjust must trade-offs between consistency and availability of the data (two of the three pillars of the **CAP Theorem**).

In a consistent mode, accurate data will always be delivered, though it may take some time. In an availability-oriented mode, data will be returned to the requester as quickly as possible, though it may not include the most recent changes (as it takes to propagate new data across different clusters).



Operational Modes

We store data in multiple namespaces with different modes

- Consistency Mode
 - Responses will be the latest
 - But it will be slow
- Available Mode
 - Responses will be fast
 - They might not be the latest

Adjust • Aerospike Meetup 02.24

Operational modes of Aerospike: Consistency and Available.

Zoom

“You will get fast responses but there’s no guarantee on the freshness of the data,” Nyavor explained, especially since Adjust writes a lot more data to disk than reads it.

There are several tools that help. Aerospike offers an **intelligent client driver** that knows which nodes on a cluster to send the requests to. The database system also allows Adjust to store secondary indexes on the speedy solid-state hard drives, an

“Aerospike does sufficiently well to be able to help us take advantage of cheaper hardware,” Nyavor said.

Overall, the system can do, on average about 1.2 million write operations per second, and 2 million get operations per second.



Aerospike operations per second at Adjust.

Zoom

About 50% of all requests take less than 500 milliseconds or less, an impressive feat given the vastness of the database itself, Nyavor said.



Aerospike operations under 500 milliseconds.

Zoom

Scanning is one of the larger operations. It is necessary to delete user records, when requested or when a customer leaves the program. Scanning an entire cluster takes about three days.

“It is a slow and intensive process because it takes a lot of resources to scan,” he said. The good news is that Aerospike can run the scan operations as a background task, temporarily suspending them when reads and writes are needed to be executed.

Database Operations

- Scans
 - 4 scan jobs each scanning a partition takes 3 days in total
 - Scanning entire cluster to remove Forget Me, Opt Out and periodic maintenance operations
 - Scans are slow intensive jobs

Adjust • Aerospike Meetup 02.24

For instance, the upgrade process is still pretty manual-intensive.

The process involves going through the change log to ensure nothing has been broken in the upgrade process.

But overall, the database is very configurable, and you need to understand all the options to get the most out of it, Nyavor said.

And if you don't know something, ask. The Aerospike support team has been really helpful in answering questions, he added.

"Don't take anything in the documentation that you don't understand for granted, because it can snowball and bite you in the ass," he said.

TNS



Joab Jackson is a senior editor for The New Stack, covering cloud native computing and system operations. He has reported on IT infrastructure and development for over 30 years, including stints at IDG and Government Computer News. Before that, he...

[Read more from Joab Jackson →](#)

Aerospike is a sponsor of The New Stack.

TNS owner Insight Partners is an investor in: Real.

SHARE THIS STORY



TRENDING STORIES

1. Timescale Hyperfunctions Ease PostgreSQL Query Complexities
2. Maria DB Gets Reactive with a Non-Blocking Connector for Java
3. 6 Tips for Better Data Science in the Cloud
- .. Distributed SQL Takes Databases to the Next Level

INSIGHTS FROM OUR SPONSOR



Aerospike is the real-time database built for infinite scale, speed, and savings. Our customers are ready for what's next with the lowest latency and the highest throughput data platform. Cloud and AI-forward, we empower leading organizations like Adobe, Airtel, Criteo, Experian, and PayPal.

[Learn More →](#)

TNS DAILY NEWSLETTER

**Receive a free roundup of the
most recent TNS articles in
your inbox each day.**

svo1975pvn1982@gmail.com

SUBSCRIBE

ARCHITECTURE

Cloud Native Ecosystem
Containers
Databases
Edge Computing
Infrastructure as Code
Linux
Microservices
Open Source
Networking
Storage

OPERATIONS

AI Operations
CI/CD
Cloud Services
DevOps
Kubernetes
Observability
Operations
Platform Engineering

THE NEW STACK

About / Contact
Sponsors
Advertise With Us
Contributions

ENGINEERING

AI
AI Engineering
API Management
Backend development
Data
Frontend Development
Large Language Models
Security
Software Development
WebAssembly

CHANNELS

Podcasts
Ebooks
Events
Webinars
Newsletter
TNS RSS Feeds



roadmap.sh

Community created roadmaps, articles, resources and journeys for developers to help you choose your path and grow in your career.

Frontend Developer Roadmap
Backend Developer Roadmap
Devops Roadmap

FOLLOW TNS



© The New Stack 2026

[Disclosures](#)

[Terms of Use](#)

[Advertising Terms & Conditions](#)

[Privacy Policy](#)

[Cookie Policy](#)